

8DM60: Multimodal AI in Biomedicine

Lecture 4: Guided self-study (BZ)

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Problem 1

A hospital builds a model to predict readmission within 30 days of discharge. Each hospital stay becomes one row: vital signs are entered as the mean of all readings during the stay. The discharge note is split into 200-word chunks; a clinical language model turns each chunk into an embedding, and the chunk embeddings are averaged into one text vector. The row and the text vector are concatenated and passed to a gradient-boosted tree classifier.

Two patients have three systolic blood pressure readings each. Patient X: 80, 100, 120 mmHg (recovering). Patient Y: 120, 100, 80 mmHg (deteriorating). Patient X's discharge note has ten chunks; one contains "new pleural effusion on today's chest film", the other nine are routine.

1. Will the model see a difference between X and Y in their blood pressure? Explain.
2. Why might the pleural-effusion sentence have little influence on X's prediction, even though the language model encodes it correctly?
3. Propose one change to the vital-sign representation and one to the text representation that would keep the lost information. For each, name one new problem it introduces.

Problem 2

A team reports a readmission model. Before splitting into training and test sets, they randomly removed non-readmitted stays until both classes were equal. On the resulting test set they report the values on the right.

In the hospital, 10% of discharged patients are readmitted within 30 days. A manager concludes: “7 out of 10 alarms will be correct in practice.”

Metric	Value
Sensitivity	0.70
Specificity	0.70
Precision	0.70
Accuracy	0.70
Area under the ROC curve (AUROC)	0.76

1. Which of the reported metrics can be expected to carry over to the hospital, and which cannot? Explain why.
2. Estimate the precision at the hospital’s real readmission rate. An approximate answer is sufficient.
3. How should the team have built its test set, and which additional result would you ask for before accepting the model?

Problem 3

A team trains a model to predict septic shock during a stay in the intensive care unit (ICU). Labels: positive if the patient developed septic shock; negative otherwise. Many negative patients received a fluid bolus shortly after early signs of sepsis appeared.

1. Explain how this labelling can bias the model's risk estimates for patients who resemble the treated negatives.
2. A student proposes removing all patients who received a fluid bolus from the training data. What does this fix, and what new problem does it create?
3. Another team keeps these patients in the training data but leaves out the treated patients who never developed shock when it measures performance. Why is it necessary to leave them out of the evaluation, and what does the reported performance then not tell you?

Problem 4

A research group has a model that reads the first 24 hours of a stay in the intensive care unit (ICU) with a bidirectional long short-term memory network (LSTM) — a sequence network that reads the data forwards and backwards — and predicts in-hospital mortality once, at hour 24. It performs well on held-out data.

The group reuses the same architecture for an hourly deterioration alert. For training, each complete ICU stay is fed as one sequence; the network outputs a risk at every hour, trained to predict deterioration within the next 12 hours. Retrospective area under the ROC curve (AUROC) is 0.95. In a live pilot on the same ward, AUROC is 0.71. A researcher concludes: “The drop is caused by dataset shift between the historical data and the pilot period.”

1. Why is the bidirectional design acceptable for the 24-hour mortality model?
2. Is the researcher’s explanation sufficient? Explain what else could cause the drop.
3. Describe how training inputs and the retrospective evaluation should be set up so that the retrospective result reflects live use.

Problem 5

Two in-hospital mortality models are evaluated on the same test set, in which 8% of patients die. Model A was trained with a class-weighted loss; Model B without.

Model	Area under the ROC curve (AUROC)	Average predicted risk	Observed deaths among patients predicted 30–40%
A	0.91	19%	12%
B	0.88	8.5%	33%

The hospital wants to compare its intensive care units (ICUs) using the ratio of observed to predicted deaths.

1. Which model does each column favour, and what does each column tell you about the model?
2. An ICU treats 200 patients whose true risk matches the hospital average, and 16 die. How many deaths would Model A predict, and what would the observed-to-predicted ratio suggest about this ICU?
3. Which model would you use for the ICU comparison, and could the other model be made suitable? What can this table alone not establish?

Problem 6

A mortality model uses structured data, clinical notes and chest X-rays from the first 24 hours. Separate combination layers are used for patients with different available data. Results at the development hospital:

Patients	Data used	Deaths	AUROC
All (10,000)	Structured	7%	0.86
With notes (7,000)	Structured + notes	7%	0.88
With a chest X-ray in the first 24 h (1,500)	Structured + notes + X-ray	14%	0.90

AUROC = area under the ROC curve. The model will be deployed at hospital K. K's data export contains no free-text notes, and K takes a chest X-ray at admission for every patient in its intensive care unit.

1. A colleague concludes that adding X-rays improves AUROC by 0.04 for all patients. Is this justified? Explain.
2. What would you expect to happen to the model's behaviour at hospital K, and why?
3. Describe what you would evaluate before using the model at K.

Problem 7

A bias audit of a mortality model gives the false-positive rates on the right (share of survivors who were flagged as high risk).

Age is among the model's strongest predictors. To make the model fairer, the team proposes to remove age from the inputs.

Age group	False-positive rate
Under 45	0.07
45–75	0.12
Over 75	0.19

1. Give a plausible reason why the model produces more false alarms in older patients.
2. Evaluate the proposal: what is it likely to change, and what is it unlikely to fix?
3. Propose an alternative approach and state what you would measure to check it.

Problem B1 (bonus)

A hospital has 60,000 admissions from 38,000 patients, recorded 2010–2023. Two teams train the same readmission model:

Team	Training data	Test data	AUROC
1	Random 80% of admissions	Remaining 20% of admissions	0.89
2	Admissions 2010–2018	Admissions 2020–2023 of patients not seen in training	0.81

AUROC = area under the ROC curve. The model will go live in 2027. Which result is the better estimate of live performance? Give two separate reasons for the gap between the teams, and describe what you would check before going live.

Problem B2 (bonus)

In a deterioration model, lactate is measured only when clinicians are concerned. Missing lactate values are replaced by the population mean, and a separate feature records whether lactate was measured. This “was measured” feature is among the three strongest predictors. Six months after deployment, the hospital introduces a protocol that measures lactate at admission for every patient.

Predict how the model’s behaviour will change, and why. Is the “was measured” feature legitimate? Propose how the hospital should handle the protocol change.